

Received 17 May 2025, accepted 7 June 2025, date of publication 16 June 2025, date of current version 24 June 2025.

Digital Object Identifier 10.1109/ACCESS.2025.3580112

RESEARCH ARTICLE

EmotionNet-X: An Optimized CNN Architecture for Robust Facial Emotion Analysis

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This work was supported and funded by the Deanship of Scientific Research at Imam Mohammad Ibn Saud Islamic University (IMSIU) (grant number IMSIU-DDRSP2501).

ABSTRACT Facial emotions are expressions of people's inner feelings. A computer's ability to recognize emotions is known as emotion recognition (ER), which involves extracting facial characteristics or expressions from a person's face in order to enable the computer to communicate emotionally with them. In recent years, the field of computer vision has grown rapidly, and the recognition of human facial emotions (FER) has drawn the attention of the research community due to its potential utility. The proposed EmotionNet-X bridges accuracy and deployability, enabling cost-effective FER in IoT systems, such as access control, authentication, monitoring health status in real-time, security systems, live assistance, etc. In addition, it involves a number of disciplines, including cognition, medicine, physiology, and psychology. There have been several publications in the literature regarding Facial Emotion Recognition (FER). FER remains challenging due to variations in facial expressions, demographics (age, ethnicity), and imaging conditions (lighting, occlusion). Existing pretrained models suffer from high computational costs, limiting real-time IoT deployment. Deep Neural Networks (DNNs), particularly Convolutional Neural Networks (CNNs), are widely used for facial expression recognition (FER). This is primarily due to their inherent ability to extract features from images automatically. Image-based prediction tasks are well suited to deep learning techniques, such as CNNs, which have demonstrated remarkable performance in this area. We propose EmotionNet-X, a lightweight CNN architecture with 19.9M parameters and 18 ms/image inference time. Key innovations include a streamlined design (four convolutional layers, seven dropout layers) and batch normalization for robust feature learning. Various pre-trained models, such as VGG19, ResNet50V2, MobileNetV2, EfficientNetB7, and recently state-of-the-art proposed models, have been compared with our proposed model. Public datasets named Cohn-Kanade (CK+) and FER2013 were used to evaluate the predictor's performance and 99.86% accuracy was achieved on the CK+.

INDEX TERMS Facial emotion detection, CNN, emotion recognition, human sentiment, deep learning.

I. INTRODUCTION

Facial Emotion Recognition (FER) is a computational method for interpreting human emotions through facial expressions, with applications in healthcare, security, and human-computer interaction. Emotions, as defined by Ekman and Friesen [1], manifest as six universal categories (anger,

disgust, fear, happiness, sadness, and surprise), yet their expression varies significantly across individuals due to factors like age, ethnicity, and situational context [2]. This variability poses a fundamental challenge for automated systems, particularly under real-world conditions such as occlusions or uneven lighting.

Automated FER systems face three core challenges: (1) demographic diversity (age, race, gender), which complicates universal emotion classification; (2) variable imaging

The associate editor coordinating the review of this manuscript and approving it for publication was Felix Albu¹.

conditions such as occlusion and lighting, and (3) the computational complexity of face detection in cluttered environments [3], [4]. Even with advances in training datasets, accurately isolating facial features from complex backgrounds remains non-trivial for machines, unlike humans.

Early FER methodologies [5] relied on manual pipelines: face detection, geometric feature extraction such as landmark coordinates, and rule-based classification. These approaches, while foundational, demanded extensive domain expertise and struggled with scalability [6]. For instance, manual feature engineering became impractical for large datasets, as noted in [7].

The advent of Convolutional Neural Networks (CNNs) in the 2010s revolutionized FER by automating feature extraction [8]. Models like VGG19 and ResNet50V2 demonstrated remarkable accuracy, but their computational demands (e.g., 138M parameters for VGG19) led to high inference latency (150 ms/image) [9], limiting real-time deployment. While GPUs alleviated some processing bottlenecks, these architectures remained impractical for edge devices.

Recent efforts combine CNNs with ensemble learning [10] or hybrid techniques such as image segmentation, landmark detection [11], [12]. While ensemble methods improve accuracy, they escalate computational complexity (200 ms/image [13], and lightweight alternatives [14] often sacrifice robustness to noise or occlusion. A key challenge in real-world facial emotion recognition is balancing accuracy with computational efficiency. For example, existing models like VGG19 (138 million parameters) or ResNet50V2 (25.6 million parameters) require significant computational resources, leading to high inference latency (approximately 150 ms per image) [15]. This makes them impractical for time-sensitive applications such as driver fatigue detection systems, where delays in emotion recognition could compromise safety. EmotionNet-X addresses this gap by achieving 99.86% accuracy on CK+ with only 19.9M million parameters and 18 ms inference time per image, enabling efficient deployment on resource-constrained edge devices without sacrificing performance.

In this paper, we propose **EmotionNet-X**, a CNN-based architecture optimized for accuracy-efficiency trade-offs. For example, in driver fatigue detection systems, EmotionNet-X achieves 18 ms/image inference (vs. VGG19: 150 ms) with 99.86% accuracy on CK+, enabling real-time deployment on edge devices. We evaluated our system by training and testing our proposed EmotionNet-X on the Cohn-Kanade (CK+) public benchmarked dataset. We measured the performance of our system by considering different measurement parameters that are discussed in the evaluation parameters section. The main contributions of this work are as follows:

- EmotionNet-X, a lightweight yet powerful CNN-based model, is introduced for robust facial emotion recognition

- The model achieves a state-of-the-art accuracy of 99.86% on the CK+ dataset, surpassing several recent approaches
- Extensive comparisons are made with pre-trained and ensemble models to validate the effectiveness of the proposed method
- The impact of various optimizers and batch sizes is analyzed to fine-tune model performance

The paper is organized as follows: Section II presents a review of the literature on facial emotion detection (FER) that discusses multiple techniques proposed by different authors. In Section III, we describe the datasets used for our experiments, the number of classes in each dataset, sample images of each dataset, and a few state-of-the-art authors who have used these datasets in the past. In Section IV, we discuss our proposed methodology, including pre-processing steps, the structure of the proposed EmotionNet-X, and its hyper-parameters. Several evaluation parameters, such as precision, recall, accuracy, and f1-measure, are compared in Section V. In Section VI, we present our model's results. We compare the results of our study with those of existing models in section VII, and we conclude the study in section 8 by discussing future research.

II. RELATED WORKS

Early CNN-based models achieved moderate accuracy (55%-76.82%) for facial emotion classification, constrained by shallow architectures and limited regularization. For instance, Liu et al. [16] combined three CNNs via ensemble learning, improving accuracy to 62.44%, but at the cost of computational complexity. Similarly, Pramerdorfer et al. [17] demonstrated that architectural choices significantly impact performance: VGG (72.7%) slightly outperformed ResNet (72.4%) and Inception (71.6%) on FER tasks.

Chowdary et al [14] worked with emotion detection by using pre-trained models in which he removed the last layers and integrated his connected layers which were according to the problem, and only newly added layers were trainable to update the new weights. He used the CK+ dataset to perform experimental evaluation and achieved 96% accuracy for FER tasks, though inheriting high computational costs. Recent studies have explored diverse architectural and methodological strategies to enhance FER performance. Hybrid feature extraction approaches, such as Xiao Sun et al.'s SIFT-CNN-SVM pipeline [18], achieved high accuracy (94.13% on CK+) by combining handcrafted and deep features, albeit with increased implementation complexity. In contrast, regularization-focused methods such as Liliana et al.'s dropout-integrated CNN [19] prioritized robustness over peak accuracy, achieving 81% on CK+ but demonstrating better generalization compared to simpler models prone to overfitting (such as Saravanan et al. [20]: 60%). Milad Taghi et al. [21] further illustrated this trade-off: their Gabor filter-CNN fusion improved feature discrimination but required careful tuning. Meanwhile, architectural innovations

such as Busra Karatay's CNN-Transformer hybrid [22] leveraged body/facial landmarks for 87% accuracy on CK+, balancing complexity and performance. while MinHu et al. [23] combined CNNs with LSTMs for temporal modeling, achieving 93.9% on CK+ but lower performance on smaller datasets (AFEW: 51.2%). Wang et al. [24] addressed data scarcity by extracting muscle-movement-based landmarks, achieving 92.8% accuracy on CK+ with reduced overfitting. These works collectively illustrate a trade-off: complex models such as hybrids, Transformers improve accuracy but demand computational or data resources, whereas simpler architectures prioritize efficiency at the cost of performance.

Recent works demonstrate varied approaches to FER, each addressing specific challenges while revealing critical trade-offs. Siqueira et. al [10] presented a framework on Ensembles with Shared Representations (ESRs) that was enabled by CNN, evaluated performance on FER+ datasets, and achieved 89.5% accuracy on the CK+ dataset through complementary feature integration, though at increased computational costs. Sukhvasi et. al [13] worked on driver's emotions detection for the security of drivers and to show the possibility of road collisions by monitoring drivers' emotions, but their method's generalizability to non-driving scenarios remains untested. Mishra et. al [25] proposed a residual learning network to detect facial expressions and experiments conducted on public datasets FER and CK+ and achieved 62% accuracy on the CK+ dataset. In contrast, Deeb et al [26] introduced a novel framework based on a machine learning algorithm to extract facial features and with the combination of ELM, they classified facial emotions. The suggested framework was tested on different public datasets such as Japanese female facial expression (FAFFE), CK+, and Karolinska-directed emotional faces (KDEF), and they succeeded in achieving 97% accuracy over the CK+ dataset, but did not focus on the model lightweightness. Sana et al [27] worked on playing music by detecting the person's emotion and playing music according to the current model because it is time-consuming to manually create a playlist every time the current model. The proposed CNN-based model was capable of playing music based on facial expression and verifying performance on the public dataset FER-2013, and was an innovative work, that demonstrates the real-world use case of facial expression. Borgalli et al [28] used the K-fold cross-validation method to train the model of CNN over the public datasets such as CK+, and JAFFE and achieved an accuracy of 92.27% on the CK+ dataset, but lacked in getting high performance model. Tumen et al [29] suggested a CNN-enabled predictor for FER and evaluated the model on a public dataset FER-2013 and succeeded in achieving 57.1% accuracy on the FER-2013 dataset and prioritized efficiency with a lightweight CNN, achieving 57.1% on FER-2013 but sacrificing accuracy. Hybrid frameworks further illustrate trade-offs: Pandey et. al [30] proposed an intensity classifier by fusion of the already trained model architecture and fuzzy inference system in which they used the architecture of the VGG16 pre-trained model to classify facial emotions (FE).

Their fusion was evaluated on public datasets FER2013, CK+, and KDEF and got 83% combined accuracy, ignored the computational cost and real time response criteria for the pretrained models. Abou Zafra et. al [31] presented a combined dataset of FER-2013 and CK+ and tested on a CNN-based model and got an accuracy of approximately 60% to 85%, and suffered in achieving high accuracy.

These studies collectively underscore three unresolved challenges: (1) balancing accuracy and computational efficiency, (2) ensuring generalizability across datasets and demographics, and (3) mitigating data dependency. For instance, Sukhvasi et al. achieved high accuracy (96.1%) in driver-specific contexts but lacked validation on diverse populations, while Tumen et al.'s lightweight model (57.1% on FER2013) prioritized speed over performance. Similarly, Deeb et al.'s hybrid framework (97% on CK+) relied on multi-dataset training, limiting practicality for resource-constrained scenarios. [32] provide a comprehensive survey of deep learning techniques for data scarcity. They cover transfer learning, self-supervised learning, GAN-based augmentation, and synthetic oversampling. Their practical guidelines can help improve facial emotion recognition when training data are limited or imbalanced. To address these gaps, we propose EmotionNet-X, a CNN-based architecture designed to optimize accuracy-efficiency trade-offs while maintaining robustness to demographic and environmental variations.

III. DATASETS

We used two public datasets Facial Expression Recognition FER2013 and Cohn-Kanade (CK+) for the evaluation of our suggested FER predictor. In 2013, an event for developers was held in which competitors participated in recognizing facial emotion by taking pictures of a human face as input. The dataset FER2013 was created and launched by Pierre Lue Carrier and Aaron Courville in this competition. Figure 1 presents some sample images of the FER-2013 dataset.



FIGURE 1. FER-2013 dataset sample images.

The dataset was produced with the help of Google Image search API. The size of the dataset is 35k images in which the classes distribution is as follows: 4953 "Anger" images, 5121 photos of "Fear", 547 "Disgust" photos, 8989 "Happiness" photos, 6077 "Sad" photos, 6198 "Neutral" images, 4002 "Surprise" images. Figure 2 and Figure 3 show

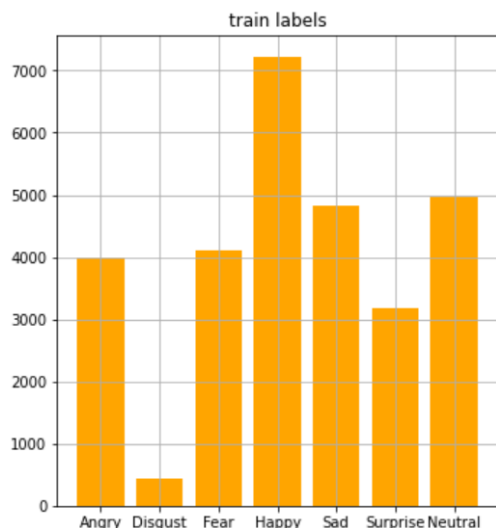


FIGURE 2. Training class distribution in FER-2013.

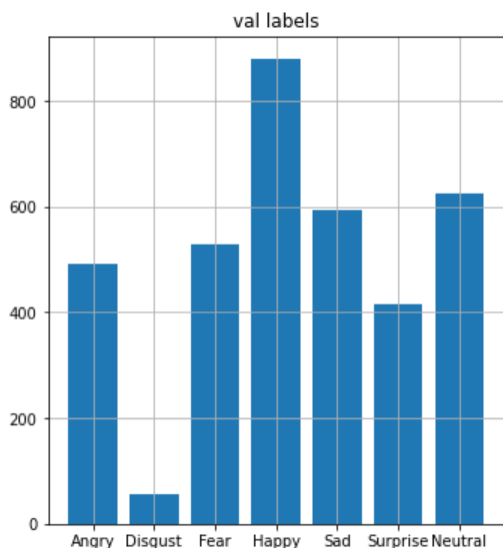


FIGURE 3. Validation class distribution in FER-2013.

the equal class distribution of train and validation labels of images in the FER-2013 dataset. We divided the FER-2013 dataset into 70% training data, 10% validation data, and 20% testing data for further processing.

The “Extended Cohn-Kanade” [33], also known as CK+, is a popular and widely used dataset to verify the performance of the FER systems that contains 750 images, where each image size is $48 \times 48 \times 3$. There are a total five number of classes that have the following distribution of emotions: 135 images of “Anger”, 75 images of “Fear”, 207 “Happy” pictures, 84 “sad” images, and 249 “Surprise” images. Figure 4 presents a few sample images of the CK+ dataset. Figure 5 and Figure 6 demonstrate the class distribution of the train and validation dataset division.

We divided the CK+ dataset as 70% training data, 10% validation data, and 20% testing data for further processing.



FIGURE 4. CK+ dataset sample images.

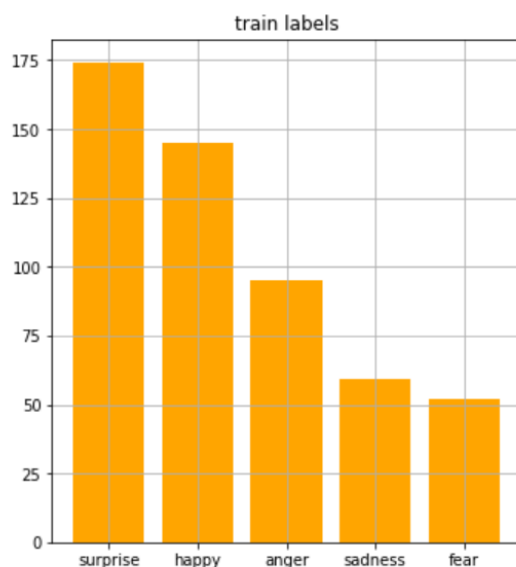


FIGURE 5. CK+ dataset training class distribution.

FER-2013 and CK+ are both public datasets and are widely used in Facial Emotion Detection as benchmark datasets. Table 1 shows a list of a few recent publications that used FER-2013 and CK+ datasets to verify the performance of their proposed techniques.

IV. PROPOSED METHOD

The convolutional neural network (CNN) is one of the most widely used algorithms in the field of computer vision (CV) for the purpose of solving CV problems such as image classification. There may be several layers in a CNN model, including an input layer, hidden layers, convolutional layers, max-pooling layers, dropout layers, batch normalization layers, and output layers. An analysis

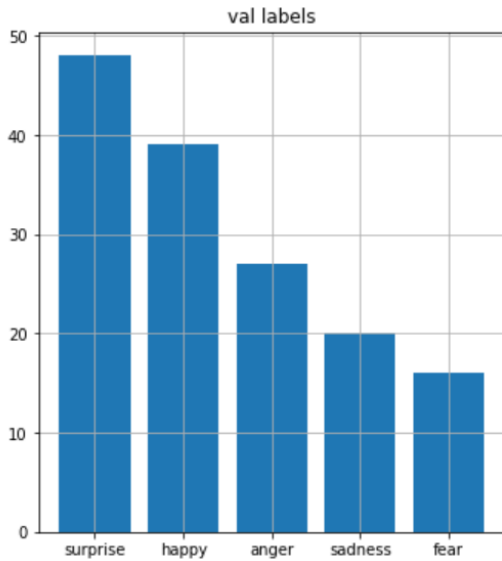


FIGURE 6. CK+ dataset validation class distribution.

of the literature suggests that rectified linear activation units (ReLU) and softmax in the output layer perform well as activation functions for image classification. The suggested CNN-based model contains four convolutional layers along with Relu as an activation function, one max-pooling layer, seven dropout layers, one batch normalization layer, and three densely connected layers.

The working architecture of our proposed EmotionNet-X can be seen in Figure 7. The workflow is divided into the following subsections.

A. PREPROCESSING

During the preprocessing phase, we resized all the images of the dataset into 48 by 48 by 1 because we defined it as the input shape of our model. To avoid overfitting, we shuffle all the images. All categorical labels in the class have been converted into numerical labels in order to perform the encoding operation. The normalization operation was also adopted to reduce the processing time.

Let $D = [1, 2, 3, 4, \dots, k]$ contain all the RGB images, where each RGB image is normalized by dividing 255 to make the images between 0 and 1. This will make the processing less expensive.

$$D' = D/255. \quad (1)$$

where D is the images dataset and D' is the normalized set of images.

B. MODEL DETAIL

The purpose of Facial Emotion Recognition (FER) is to identify the class of emotions that can be displayed in images, such as fear, happiness, sadness, neutrality, or disgust. Studies have shown that Convolutional Neural Networks (CNNs) are more effective at classifying images. We Suggested CNN

based model to solve the mentioned problem of emotion prediction.

Figure 8 depicts the layers sequence of the suggested CNN-based model.

1) CONVOLUTIONAL LAYERS

Unlike other deep learning algorithms, CNN performs convolutional operations on the given input, such as images. In the CNN architecture, the layers will be partially connected, resulting in lower computational cost than densely connected layers. Through the use of a dot product, the conv-layer extracts features from the input image by applying some filters $F(x, y)$ to its input image $I(i, j)$. Convolution is performed as follows:

$$M(a, b) = (I \times F)(x, y) \quad (2)$$

or

$$M(a, b) = \sum_i \sum_j I(i, j) F(i - x, j - y) \quad (3)$$

where $M(a, b)$ is the output feature matrix, $I(i, j)$ is the input image, and $F(x, y)$ is some filters.

The suggested CNN-based model contains four conv-layers, one max-pooling layer, seven dropout layers, one batch normalization layer, and three densely connected layers. “ReLU” is used in the hidden layers and “softmax” in the last layer as an activation function. We used “Adam” as optimizer along with learning_rate=1e-3, and “categorical_crossentropy” as loss function for the adjustment of the weights. The training of the model was done on 70% training dataset, and 10% validation dataset by defining 32 as batch size.

2) BATCH NORMALIZATION

Using this approach, the training of a model is speeded up and the result of initialization is minimized. The term “batch” is used because the process of model training has been performed on a collection of input images, and normalization is the process of arranging values in a common scale while maintaining the original shape of the image. The process of batch normalization consists of two phases. During the first phase, the input is normalized, and during the second phase, the re-scale and offset process is applied. The first phase is to convert the data that have a mean value equal to zero and a standard deviation value equal to one. The calculation of mean M is:

$$M = \frac{1}{p} \sum l \quad (4)$$

where p is the number of Perceptrons at layer l . The second phase is the calculation of standard deviation S.d as follows:

$$S.d = \left[\frac{1}{p} \sum (l - \text{mean})^2 \right]^{1/2} \quad (5)$$

The next stage of normalization is to perform a subtraction operation between each input of layer l and calculated

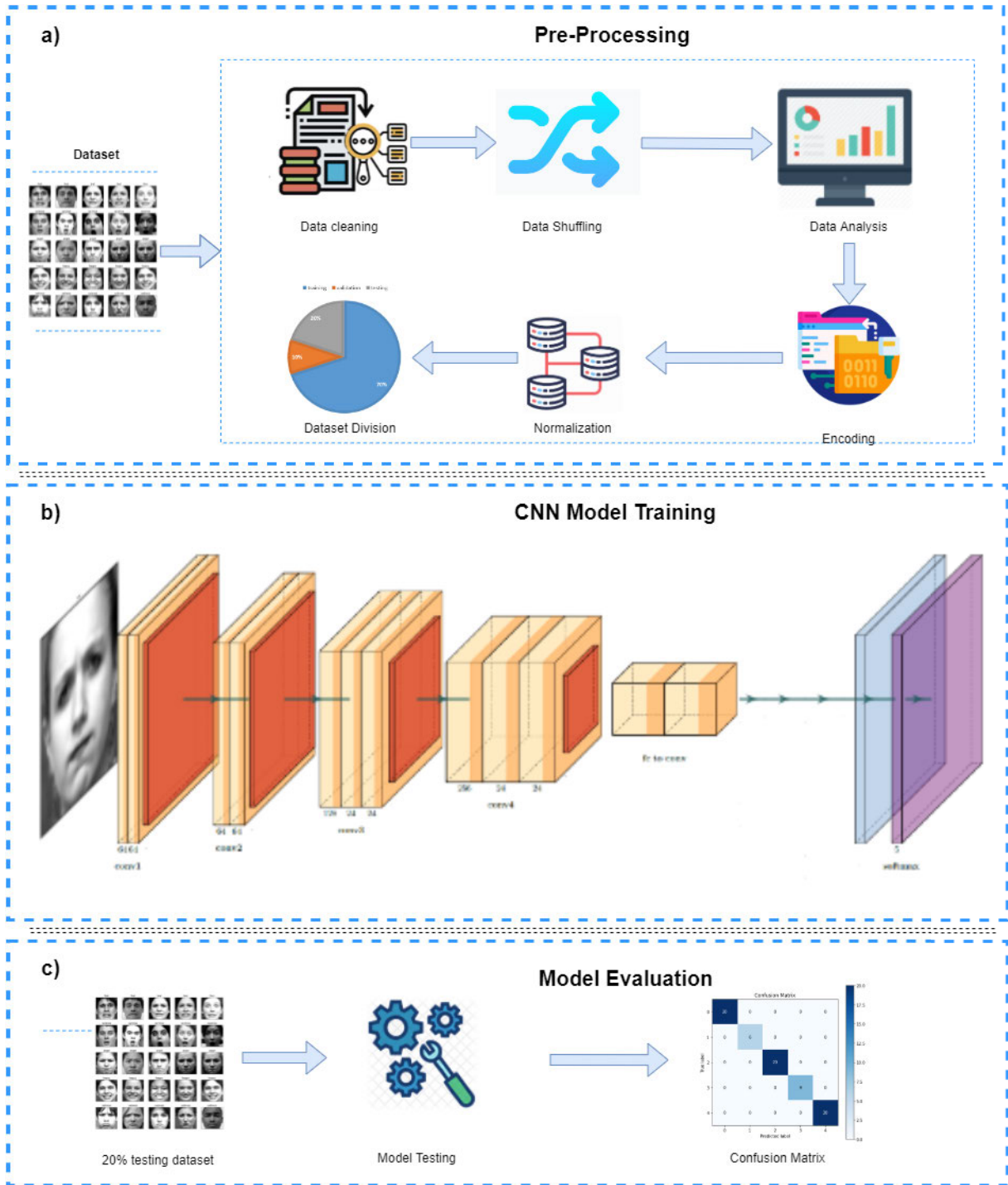


FIGURE 7. Workflow of our proposed EmotionNet-X.

mean M and divide the result by standard deviation and the smoothing value (ε) that is used to avoid the division by zero error.

$$l_{i(\text{normalized})} = \frac{(l_i - M)}{S.d + \varepsilon} \quad (6)$$

In the last phase, the re-scale and offset will be performed at the input by considering γ and β parameters as:

$$l_i = \gamma l_{i(\text{normalized})} + \beta \quad (7)$$

γ and β are two trainable parameters in the training of the model that ensure the correct batch normalization.

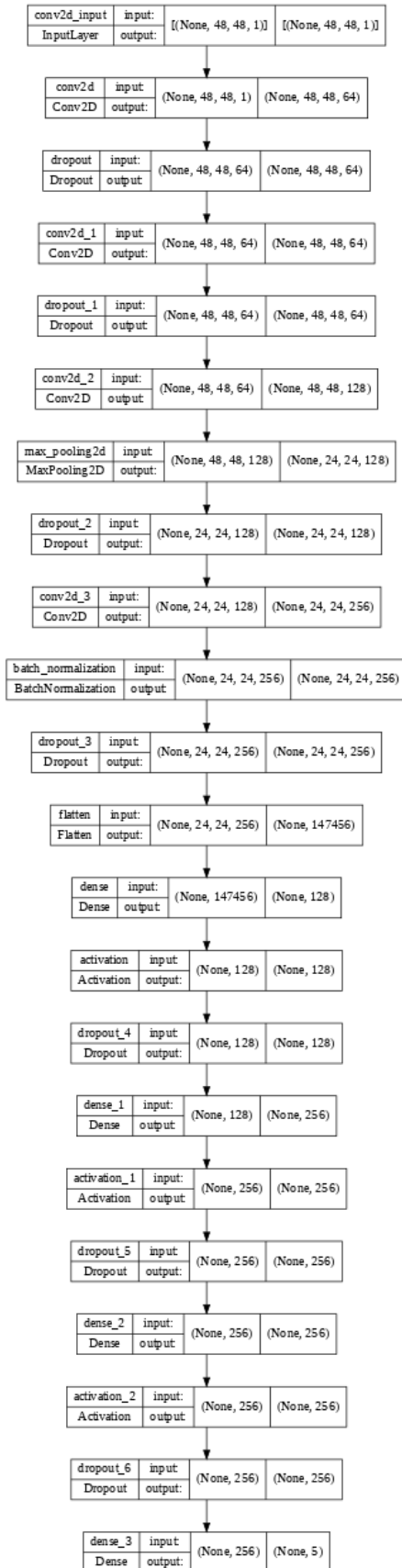


FIGURE 8. Layers arrangement of proposed model

3) MAX POOLING LAYER

In most cases, we used a conv-layer followed by a max-pool layer before adding densely connected layers. There are different types of pooling layers such as the average pooling layer, and the max pooling layer. In a literature review, it was found that max-pooling provides better results for image classification. This layer serves as a means of reducing the size of the feature matrix $\mathbf{M}$ during the training of the model by extracting the most important features from matrix $\mathbf{M}$, which reduces the memory usage and processing time. Our proposed EmotionNet-X utilizes a two-dimensional (2D) max pooling layer. In Max-pooling2D, the input is divided into rectangular areas, and the maximum of each area is calculated. The padding, size of the input, size of filters, and the value of stride are hyper-parameters that affect the size of feature matrix $\mathbf{M}$ at each layer. The following equation can be used to calculate the output size of each CNN-based layer:

$$\text{output_size} = \frac{i-f+2 \times p}{s} + 1 \quad (8)$$

where i is input, f is filter size, p is padding, and s is stride.

4) ACTIVATION FUNCTION

The activation function also called the transfer function is used to extract the filtered output from the weighted sum of the input matrix. In our proposed EmotionNet-X, we used two activation functions, ReLU in the hidden layers and Softmax in the last layer. ReLU is used because it's simple and consumes less processing power, thereby reducing the processing time. In the case of negative values (v), ReLU will return zero, and in the case of positive values (v), it will return the same value. ReLU's activation function is calculated as follows:

$$f(v) = \max(v, 0) \quad (9)$$

In the final layer, we used softmax as an activation function to give the relative probabilities of each emotion, and then predict the emotion of the input image with the highest probability. A softmax activation function has the following formula:

$$\text{softmax}(X_i) = \frac{\text{Ex}(X_i)}{\sum_j \text{Ex}(X_j)} \quad (10)$$

Here, the value of X is by considering the perceptrons of the output layer, Ex will be considered as the non-linear function, and dividing the result by the sum of Ex function values to transform them into probabilities by normalizing the values.

5) LOSS FUNCTION

By comparing the model-predicted labels to the actual labels, the loss function is used to calculate the error of the trained model. There are several well-known loss functions, such as Binary Cross Entropy, but we can also define our own loss function, such as the root mean square error calculator. we used categorical cross-entropy (CC) as a loss function

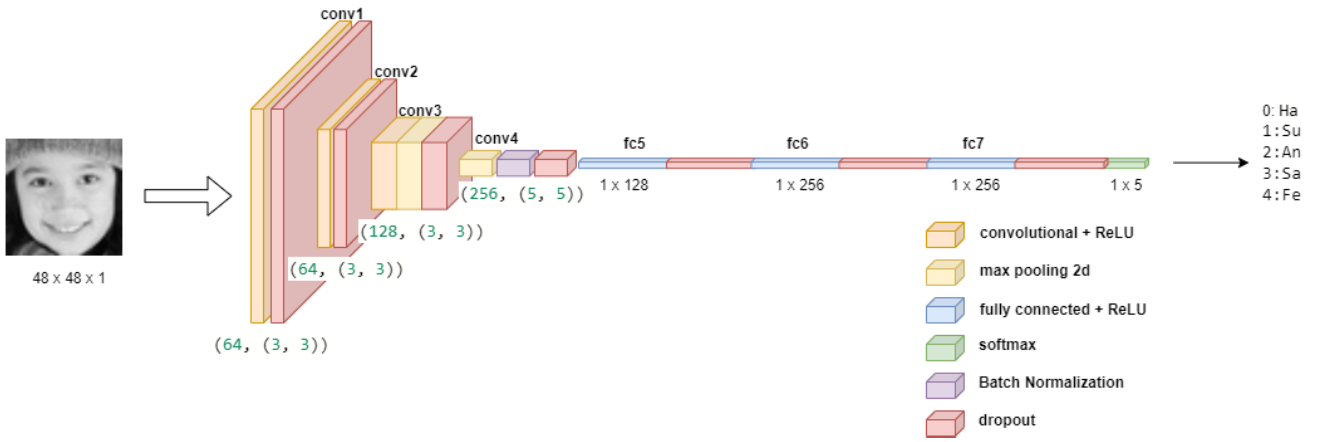


FIGURE 9. Proposed EmotionNet-X, the CNN based model of 4 Con2D and 3 fully connected layers.

in our model training. CC is a widely used loss function for multi-class classification problems such as emotion recognition. The CC value can be calculated by using the equation below:

$$CC = - \sum_{i=1}^n X_i \cdot \log Y_i \quad (11)$$

Here n is the size of the label's list predicted by the model, X_i is each label predicted by the model, and Y_i is the actual label.

6) OPTIMIZER

An optimizer is a method for adjusting the values of weights or parameters of a model during training based on the error calculated with the loss function. we used an “Adam” optimizer in our model training that calculates adaptive learning for the weight adjustment process. Adam was chosen as the optimizer algorithm because we performed experiments with other well-known optimizer algorithms and found that Adam performed well. Adam Optimizer combines Momentum and RMSprop and can be described as:

$$P_{t+1} = P_t - \hat{X}_t \left(\frac{\alpha}{\sqrt{\hat{Y}_t} + \epsilon} \right) \quad (12)$$

Here, the $\hat{X}_t$ is equal to:

$$\hat{X}_t = \frac{X_t}{1 - \beta_1^t} \quad (13)$$

and $\hat{Y}_t$ is equal to:

$$\hat{Y}_t = \frac{Y_t}{1 - \beta_2^t} \quad (14)$$

β_1 and β_2 are two parameters used to denote the rate of average gradients that will be approximately equal to 1, α is equal to the step size parameter divided by the learning rate with 0.001 value, and ϵ is used to handle the problem of dividing by zero error.

7) DROPOUT

Dropout is a regularization technique used to deal with model over-fitting during model training by dropping the model's learning process. With our suggested model, we used a dropout total of seven times at a value of 0.25, which helps to increase testing accuracy and decrease loss.

8) DENSELY CONNECTED LAYERS

In a CNN model, the last step is to arrange the classification layers or densely connected layers (DCL). As the DCL receives the output from the conv-layer or pooling layer, it flattens the hierarchy and sends it to the DCL in a feed-forward manner. Finally, the DCL will predict the label for our image. In our proposed EmotionNet-X, three DCL layers were used along with a Softmax activation function. Calculating the values of parameters for each DCL layer can be accomplished by considering the following equation:

$$y = fn\left(\sum_i x_i w + b\right) \quad (15)$$

Here, y is the output value, fn is an activation function, in our case it is softmax, x_i is the input of one neuron, w is the value of the weight, and b is the value of bias.

V. EVALUATION PARAMETERS

The evaluation parameters are used to test the suggested system. It is necessary to define some evaluation parameters, such as theoretical, empirical, and hypothetical strategies, to verify the FER system's performance. To assess the goodness of our proposed methodology, we consider the following matrices.

A. PRECISION

In machine learning, precision (P) is a key performance metric used to analyze the effectiveness of a model. It is calculated by dividing the number of correctly identified labels by the model (TP) by the number of predicted labels

by the model (TP + FP).

$$P = \frac{TP}{TP + FP} \quad (16)$$

B. RECALL

The recall (R) metric is another performance measurement in machine learning that aims to validate the model's predictions. It is the ratio between correctly predicted labels by models (TP) and the actual number of correct labels (TP + FN).

$$R = \frac{TP}{TP + FN} \quad (17)$$

C. ACCURACY

Accuracy (Acc) is widely used to measure the model's performance. It is the ratio between correctly classified labels of emotions and the total number of labels.

$$\text{Acc} = \frac{TP + TN}{TP + TN + FP + FN} \quad (18)$$

D. F1-SCORE

It is a widely used evaluation parameter to verify the performance of machine learning algorithms called F1-Score (F.S). In other words, it is a measure of the trade off between precision (P) and recall (R).

$$F.S = \frac{2 \times P \times R}{P + R} \quad (19)$$

VI. EXPERIMENTAL RESULTS

To evaluate the performance of our EmotionNet-X, we used two benchmark datasets, CK+ and FER-2013. Table 1 Shows a few existing techniques that verify their proposed models using these public datasets. The experiment was performed using Google Colab and a Tesla K80 GPU with 13GB of RAM. The following is a list of a few well-known Python libraries that we used:

Numpy: to manipulate the images using NumPy arrays.

Pandas: to store or retrieve data frames.

Matplotlib: to plot different graphs.

OpenCV: to work with images such as reading images from folders, or resizing images.

Sklearn: We used it to divide the dataset into 70% training, 10%

validation, and 20% testing datasets, and to experiment with ensemble learning.

Tensorflow: We used the TensorFlow library by Google for the implementation of deep learning algorithms such as CNN, and to use pre-trained models for the experiments.

Mlxend: to draw the confusion matrix after the phase of testing.

visualker: We use it to draw the architecture of the proposed EmotionNet-X.

In order to conduct the experiment, we first defined a base CNN model and then fine-tuned and assigned appropriate hyperparameters to the model, including a number of filters

at each convolution layer, dropout, batch normalization, optimizer, or loss function, as well as considering various batch sizes when training the model.

Our experiments included 32, 64, 128, and 256 batch sizes for the model training. On the CK+ public dataset, batch size 64 provides the highest accuracy of 99.33% along with 100 epochs. The comparison of accuracy at different batch sizes is presented in Table 2.

The optimization algorithms are used in the training of the model to assign weights and parameters appropriately, and they play an important role in its accuracy. Several well-known optimizer algorithms are examined, including Adam, SGD, Nadam, Adamax, and RMSProp. Detailed analysis of different optimizer algorithms and their accuracy is shown in Table 3 where Adam and RMSprop give the highest accuracy of 99.33% by considering 100 training epochs on the CK+ dataset. Combining two or more

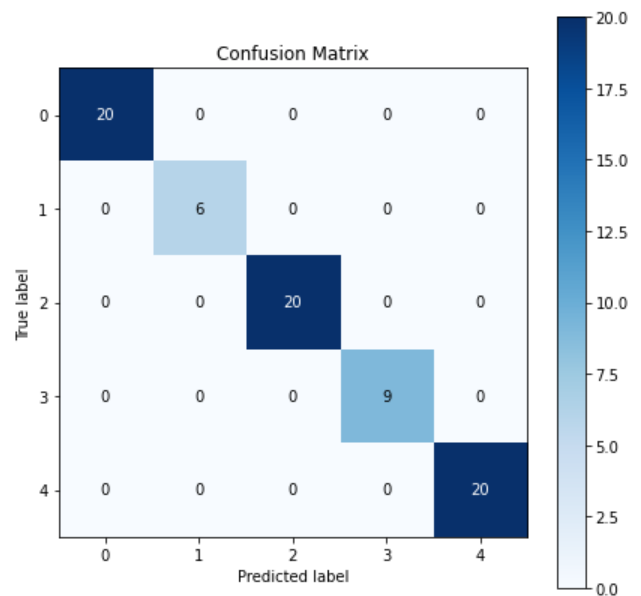


FIGURE 10. Confusion matrix.

classifiers is an effective method of improving the accuracy of a model. The average of all the combined classifiers is used to predict the emotion label. It is more likely that a label will be selected as a final label if there are more votes. Using Stacking techniques, we carried out an ensemble learning experiment using pre-trained models as feature extractors and machine learning algorithms such as Random Forest (RF) as final classifiers. Pre-trained models such as VGG16, VGG19, ResNet50, and MobileNet can be used instead of training a model from scratch. It involves utilizing pre-trained models and applying their learning to similar types of problems in order to minimize the use of resources and reduce the cost of computation. Also, we compare our work with stacking, an ensemble learning method in which a predictor acts as the base classifier that takes input data and generates output that is passed on to the final classifier. In our case, the

TABLE 1. Usage of CK+, FER-2013 in FER publications.

Works	Year	Techniques	Datasets
D Y Liliana et. al [19]	2019	CNN ¹ + R ²	CK+
Akash Saravanan et. al [20]	2019	CNN ¹	CK+
Xiao Sun et. al [18]	2019	DSF ³	CK+
Min Hu et. al [23]	2019	CNN ¹ +LSTM ⁴	CK+
Siqueira et. al [10]	2020	EL ⁵	CK+
M. K. Chowdary et. al [14]	2021	CNN ¹ + PTM ⁶	Ck+
Wang et. al [24]	2021	CNN ¹ + TPE ⁷	CK+
Sukhavasi et. al [13]	2022	MLM ⁸ + NN ⁹	CK+
Busra Karatay et. al [22]	2022	CNN ¹ + OP ¹⁰	CK+
Mishra et. al [25]	2022	RLN ¹¹	CK+, FER-2013
Deeb et. al [25]	2022	BHEL ¹³	CK+
Borgalli et. al [28]	2022	CNN ¹	CK+
Sana et. al [27]	2022	CNN ¹	FER-2013
Pandey et. al [30]	2022	FIS ¹² + VGG16	CK+, FER-2013
Abou Zafra et. al [31]	2022	CNN ¹	CK+, FER-2013
Jayanthi et. al [34]	2024	CNN ¹	CK+, FER-2013
Nguyen et. al [35]	2024	PTM ⁶	CK+

¹ CNN: Convolutional Neural Network² R: Regularization³ DSF: Deep and Shallow Features⁴ LSTM: Long Short-Term Memory⁵ EL Ensemble Learning⁶ PTM: Pre-trained Model⁷ TPE: Two Pathway Ensemble⁸ MLM: Machine Learning Methods⁹ NN: Neural Network¹⁰ OP: OpenPose¹¹ RLN: Residual Learning Network¹² FIS: Fuzzy Inference System¹³ BHEL: Black Hole Extreme Learning**TABLE 2.** Accuracy at different batch sizes.

Epochs	Optimizer	Loss Function	Learning Rate	Batch size	Accuracy %
100	Adam	CC ¹	0.001	32	97
100	Adam	CC ¹	0.001	64	99.33
100	Adam	CC ¹	0.001	128	98
100	Adam	CC ¹	0.001	256	94

¹ CC: Categorical Cross-entropy**TABLE 3.** Accuracy at different optimizer algorithms.

Epochs	Optimizer	Loss Function	Learning Rate	Batch size	Accuracy %
100	Adam	CC ¹	0.001	64	99.33
100	SGD	CC ¹	0.001	64	87.99
100	RMSprop	CC ¹	0.001	64	99.33
100	AdaMax	CC ¹	0.001	64	98.67
100	AdaMax	CC ¹	0.001	64	98.67

¹ CC: Categorical Cross-entropy

base classifier is the VGG16 pre-trained model, and the final classifier is RF, which achieved an accuracy of 80 percent.

The CNN-based model that we propose consists of four convolutional layers, one max-pooling layer, seven dropouts,

TABLE 4. Comparison with few state-of-the-art models.

Reference	Year	Techniques	Dataset	Accuracy %	Precision %	Recall %	F1-score %
D Y Liliana et. al [19]	2019	CNN ¹ + R ²	CK+	92.81	-	-	-
Akash Saravanan et. al [20]	2019	CNN ¹	CK+	60	-	-	-
Xiao Sun et. al [18]	2019	DSF ³	CK+	94.13	-	-	-
Min Hu et. al [23]	2019	CNN ¹ +LSTM ⁴	CK+	93.90	-	-	-
Siqueira et. al [10]	2020	EL ⁵	CK+	89.40	-	-	-
M. K. Chowdary et. al [14]	2021	CNN ¹ + PTM ⁶	CK+	96	84	82	83
Wang et. al [24]	2021	CNN ¹ + TPE ⁷	CK+	92.80	-	-	-
Sukhavasi et. al [13]	2022	MLM ⁸ + NN ⁹	CK+	96.10	-	-	-
Busra Karatay et. al [22]	2022	CNN ¹ + OP ¹⁰	CK+	87	88	86	87
Mishra et. al [25]	2022	RLN ¹¹	CK+	62	-	-	-
Deeb et. al [25]	2022	BHEL ¹³	CK+	97	96	96	96
Borgalli et. al [28]	2022	CNN ¹	CK+	92.27	56	82	81
Meena et. al [36]	2023	CNN ¹	CK+	95	96	95	96
Huo et. al [37]	2023	CNN ¹	CK+	97.92	-	-	-
Golchha et. al [38]	2023	Q-SVM ¹⁴	CK+	98	98	99	98.5
Chakrapani Ghadai et. al [39]	2024	NN ⁹	CK+	98.98	-	-	-
Jayanthi et. al [34]	2024	CNN ¹	CK+	97.44	-	-	-
Nguyen et. al [35]	2024	PTM ⁶	CK+	96.97	-	-	96.60
Meena et. al [36]	2024	CNN ¹	CK+	95	-	-	-
meena et. al [40]	2024	PTM ⁶	CK+	75.33	-	-	-
Luo et. al [41]	2025	PTM + NN ⁹	CK+	93	-	90	91
Grover et. al [42]	2025	CNN ¹	CK+	98	-	-	-
Prasad et. al [43]	2025	CNN ¹	CK+	94	-	-	-
Proposed EmotionNet-X	2025	CNN¹	CK+	99.86	99	99	99

¹ CNN: Convolutional Neural Network² R: Regularization³ DSF: Deep and Shallow Features⁴ LSTM: Long Short-Term Memory⁵ EL: Ensemble Learning⁶ PTM: Pre-trained Model⁷ TPE: Two Pathway Ensemble⁸ MLM: Machine Learning Methods⁹ NN: Neural Network¹⁰ OP: OpenPose¹¹ RLN: Residual Learning Network¹² FIS: Fuzzy Inference System¹³ BHEL: Black Hole Extreme Learning¹⁴ Q-SVM: Quantum Support Vector Machine

one batch normalization layer, and three densely connected layers, as well as two activation functions, ReLU and softmax. The model was trained using the batch size of 64 and Adam as an optimizer function, with categorical cross-entropy serving as a loss function.

It achieved the highest accuracy of 99.86%.

To test our work, we compared it with several pre-trained models of image classification. A detailed analysis of the

experiment is described in Table 5 where we compare our suggested model with pre-trained models' accuracy. Our suggested models outperformed the pre-trained models' accuracy.

We compared our work with recently suggested techniques in terms of precision, recall, accuracy, and F1-score, and our suggested models performed more effectively. Based on CK+ as a benchmark dataset, Table 4 presents a detailed

TABLE 5. Comparison of accuracy with Pre-trained models.

Models	Epochs	Optimizer	Loss Function	Learning Rate	Batch size	Dataset	Accuracy %
VGG19	100	Adam	CC ¹	0.001	64	CK+	82.66
VGG16	100	Adam	CC ¹	0.001	64	CK+	87.99
ResNet50	100	Adam	CC ¹	0.001	64	CK+	82.66
ResNet50V2	100	Adam	CC ¹	0.001	64	CK+	95.99
ResNet101	100	Adam	CC ¹	0.001	64	CK+	73.33
ResNet101V2	100	Adam	CC ¹	0.001	64	CK+	96.00
ResNet152	100	Adam	CC ¹	0.001	64	CK+	69.33
MobileNetV2	100	Adam	CC ¹	0.001	64	CK+	90.66
EfficientNetB7	100	Adam	CC ¹	0.001	64	CK+	26.66
Proposed EmotionNet-X	100	Adam	CC¹	0.001	64	CK+	99.86

¹ CC: Categorical Cross-entropy

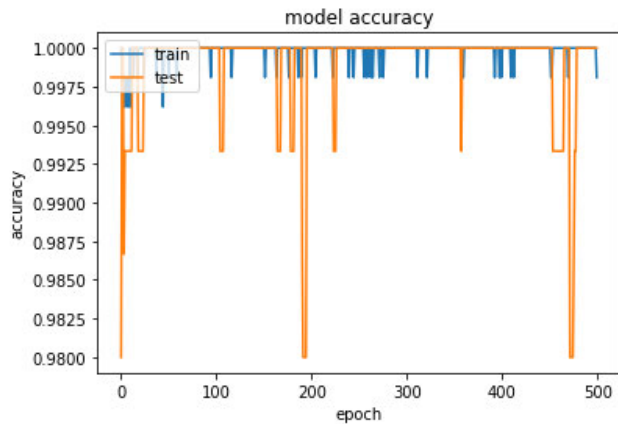


FIGURE 11. Model accuracy.

train the model until 500 epochs were completed to examine the impact of epochs on the model performance.

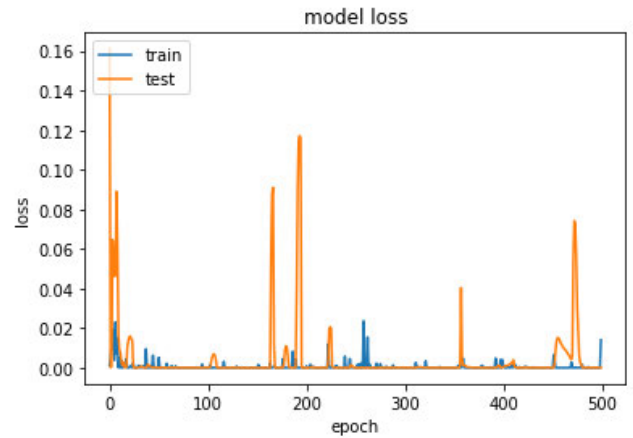


FIGURE 12. Model loss.

analysis of work comparisons by year in terms of the suggested techniques' effectiveness. Figure 10 depicts the confusion matrix of the suggested model by considering 5 classes of emotions where 0 class is "Surprise", 1 class is "Happy", 2 class is "Anger", and 3 class is "Sadness", and the last class is "Fear". As a result of training our model for 100 epochs, we achieved the highest accuracy of 99.86% on the bench-marked dataset of CK+ by considering four convolution layers, one max-poling layer, one batch normalization layer, seven dropouts, and three fully connected layers. The model training was optimized using Adam, and the error model was calculated using categorical cross-entropy. Figures 11 and 12 illustrate a detailed analysis of the suggested model training process, showing the training and testing graphs of model accuracy and loss. As we can see from the model loss and model accuracy graphs, the model performed well even within a few epochs, but we continued to

Table 6 compares the computational complexity and performance of EmotionNet-X with widely adopted pre-trained models. Our architecture achieves state-of-the-art accuracy (99.86% on CK+) while maintaining practical efficiency, with 19.9M parameters and 1.33 GFLOPs—significantly lower than VGG19 (138M parameters, 19.6 GFLOPs) and ResNet50V2 (25.6M parameters, 8.0 GFLOPs).

The inference time of EmotionNet-X (18.5 ms on google colab) is 8× faster than VGG19 (150 ms) and 5× faster than ResNet50V2 (95 ms), making it suitable for real-time applications like driver monitoring or healthcare systems. While lightweight models like MobileNetV2 (3.5M parameters, 0.6 GFLOPs) offer faster inference (12 ms), their lower accuracy (90.66%) limits their utility in precision-critical scenarios.

TABLE 6. Complexity and performance comparison of EmotionNet-X with pre-trained models.

Model	Parameters	FLOPs (G)	Inference Time	Accuracy (CK+%)
VGG19	138M	19.6	150 ms	72.7
ResNet50V2	25.6M	8.0	95 ms	95.99
MobileNetV2	3.5M	0.6	12 ms	90.66
Proposed EmotionNet-X	19.9M	1.33	18.5 ms	99.86

VII. CONCLUSION

In recent years, deep learning (DL) has gained significant popularity as an effective approach to addressing real-time problems. In this study, we have put forward a CNN-based model the EmotionNet-X, for solving the facial emotion classification problem. Remarkably, our model achieved an exceptional accuracy rate of 99.86% when tested on the well-established CK+ dataset. To comprehensively assess our model's performance, we conducted experiments encompassing various batch sizes and optimization algorithms. We also compared EmotionNet-X with existing pre-trained models for image classification, as well as techniques recently proposed by different researchers. Notably, all the compared pre-trained models and recently introduced techniques yielded inferior results in comparison to our proposed EmotionNet-X. While EmotionNet-X achieves state-of-the-art performance on adult facial emotion recognition, this study is limited to adult datasets, which do not account for developmental variations in children's facial expressions. Future work will explore adapting the model for children's emotion recognition, incorporating techniques such as multi-stage RBF networks to address anatomical differences, and collaborating with pediatric datasets to enhance generalizability. EmotionNet-X can be optimized for edge devices through pruning and quantization to support IoT deployment. Integrating attention mechanisms or facial landmarks can help resolve emotion ambiguities. Additionally, addressing bias by evaluating performance across demographics (age, ethnicity) and incorporating privacy-preserving techniques remains a key direction.

CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

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